



BRAC UNIVERSITY
Department of Computer Science Engineering
MIDTERM EXAMINATION, FALL 2022
CSE 350: Digital Electronics and Pulse Technique



Total Marks: **60 (20 x 3)**

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet
- Use activation voltage $V_\gamma(\text{diode}) = 0.6 \text{ V}$, $V_\gamma(\text{transistor}) = 0.5 \text{ V}$, $V_D = 0.7 \text{ V}$, $V_{BE}(\text{forward active}) = 0.7 \text{ V}$, $V_{BE}(\text{sat}) = 0.8 \text{ V}$ and $V_{CE}(\text{sat}) = 0.1 \text{ V}$ for all the questions.
- There is no need to draw circuits in your answer sheet.

Name:	ID:	Section:
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Question 1

For the DTL circuit in **figure 1** assume that $\beta_F = 40$. [hint: The Q_1 transistor can only operate either in saturation or cutoff mode] [20]

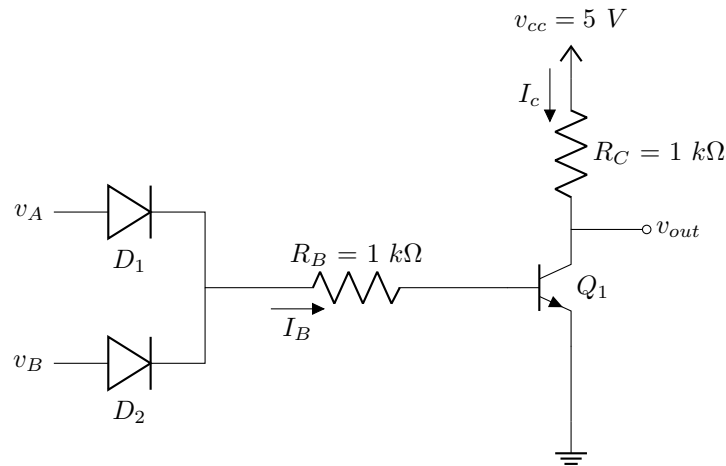


Figure 1

CO2	(a)	Assuming $v_A = v_B = 5 \text{ V}$ on the inputs, find I_C and I_B	[4]
CO1	(b)	In case of (a), identify the operating modes of D_1 , D_2 and Q_1 .	[3]
CO2	(c)	Calculate power dissipation of the total circuit in (a)	[4]
CO1	(d)	Assuming $v_A = v_B = 0.1 \text{ V}$ on the inputs, determine the operating modes of D_1 , D_2 and Q_1 .	[3]
CO2	(e)	Mention which logic gate the circuit represents. Draw the truth table for that logic gate.	[6]

Question 2

For the DTL circuit in **figure 2**, the parameters are $\beta_F = 25$, $V_{OH} = 11.5\text{ V}$ and $V_{OL} = 0.1\text{ V}$. [20]

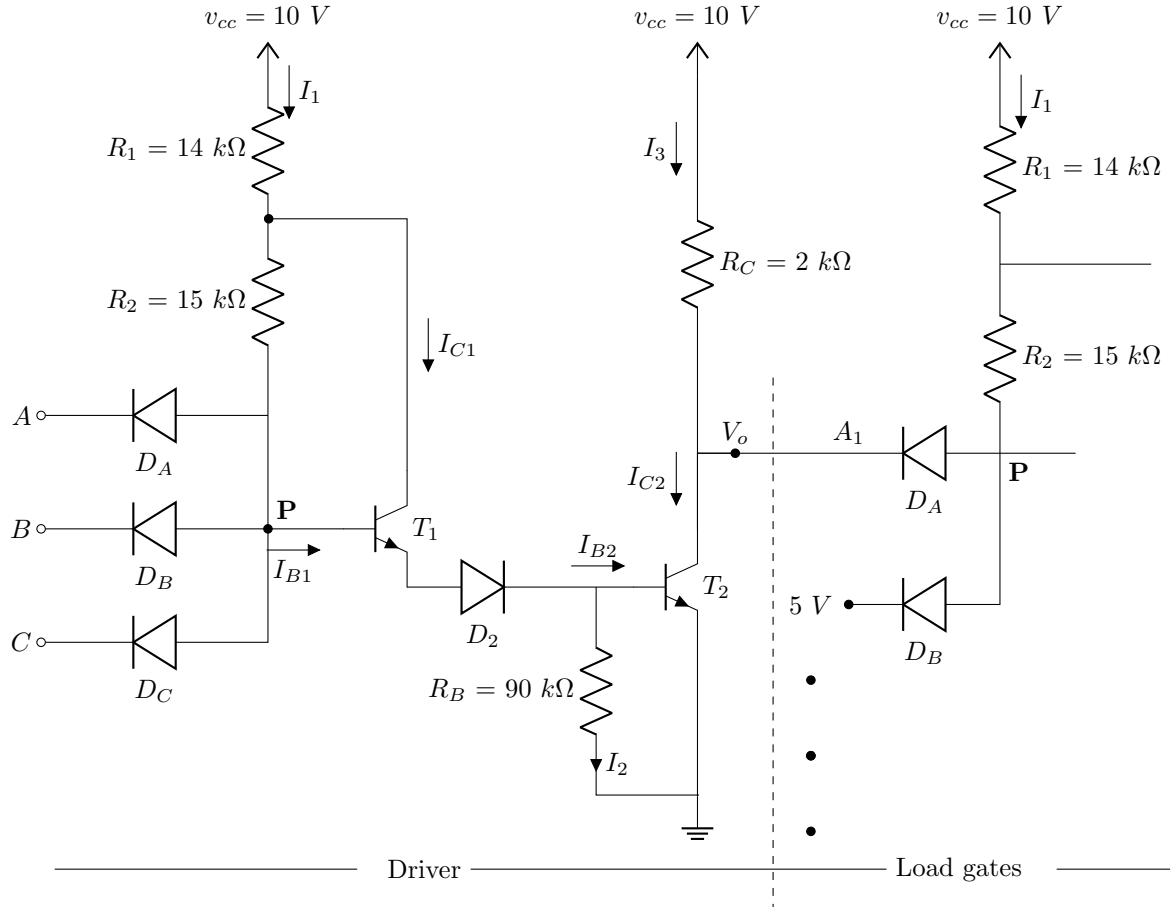


Figure 2

CO1	(a)	If all inputs of the circuit are HIGH , estimate the magnitude of the noise voltage (in Volt) which will cause the gate to malfunction?	[8]
CO2	(b)	Calculate the value of I_1 , I_{B2} , I_3 in Driver circuit and I_1 in a Load circuit when all inputs of driver are HIGH .	[8]
CO2	(c)	Suppose, we want maximum possible fanout to be 20 when all inputs are HIGH . Choose a suitable value of R_C to ensure this	[4]

Question 3

For the given totem pole TTL circuit in **figure 3**, common emitter current gain, $\beta_F = 40$ and reverse common emitter current gain of the transistors $\beta_R = 0.4$. Assume **both inputs** are logical low or **0.1 V** [20] .

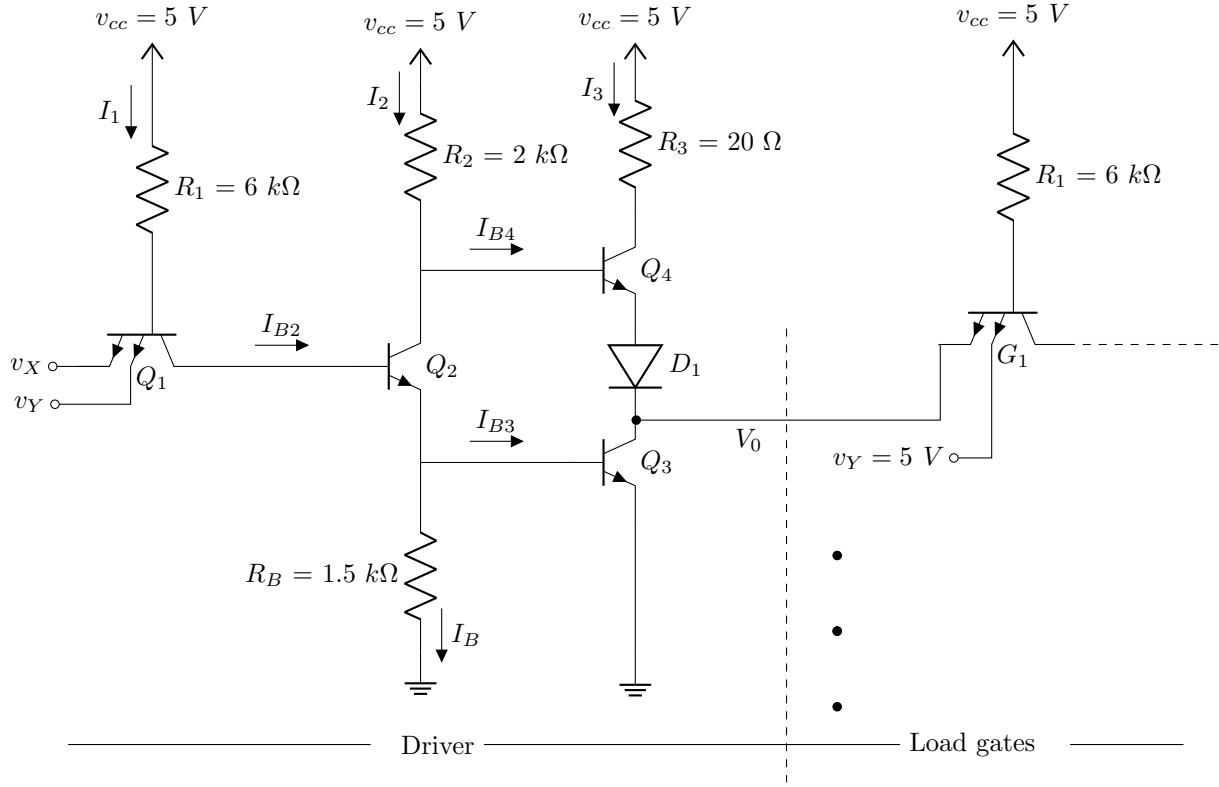


Figure 3

CO2	(a)	For fanout of $N = 8$, measure the output voltage V_o	[8]
CO1	(b)	Using the data of (a) verify the operating mode of Q_4 transistor mathematically. (<i>hint: for active mode $V_{ce} > 0.1\text{ V}$</i>).	[6]
CO2	(c)	Calculate the power dissipation of the driver for fanout of $N = 8$? [<i>hint: use the data from (a)</i>]	[6]

Question 4

For the Diode logic **AND** gate in **figure 4** consider, $V_{HIGH} = 5\text{ V}$ and $V_{LOW} = 0\text{ V}$ at the inputs [20] .

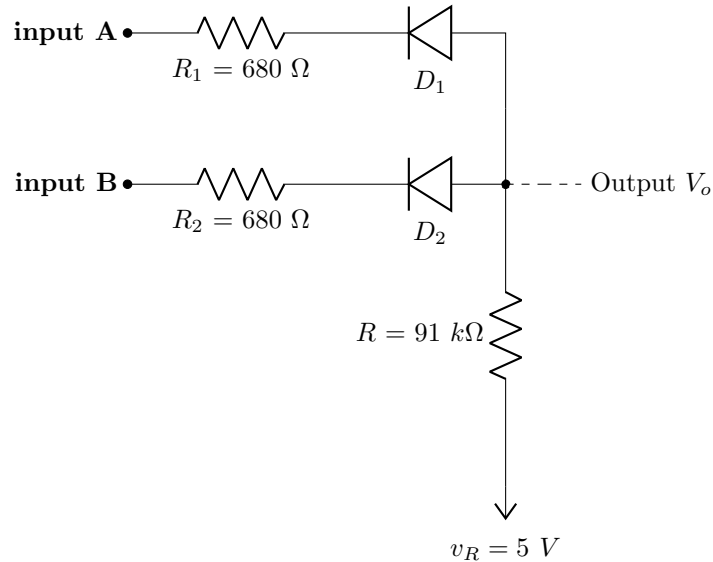
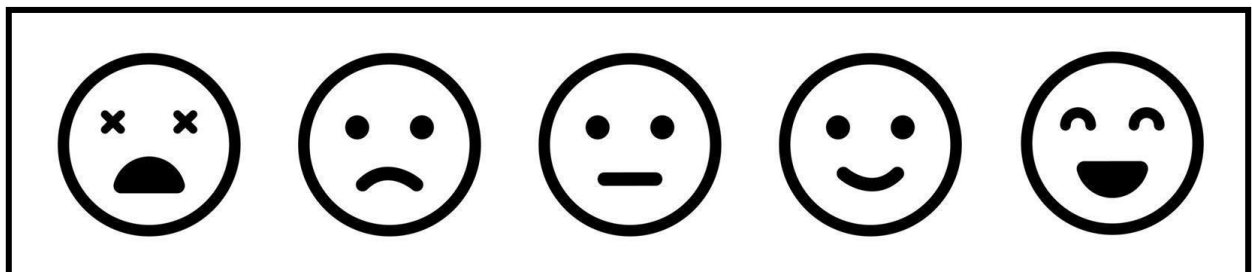


Figure 4

CO2	(a)	Calculate the output voltage V_o of this logic gate circuit for all input combinations and verify boolean AND operation..	[8]
CO2	(b)	Find the maximum power dissipation of this circuit in mW .	[6]
CO2	(c)	Calculate the minimum value of R in KΩ , if we want to keep the maximum power dissipation under 1mW . [hint: You may neglect the value of R_1 or R_2 as they are very small compared to R .	[6]

After you are done with the exam, please indicate where you stand on the smiley face spectrum.





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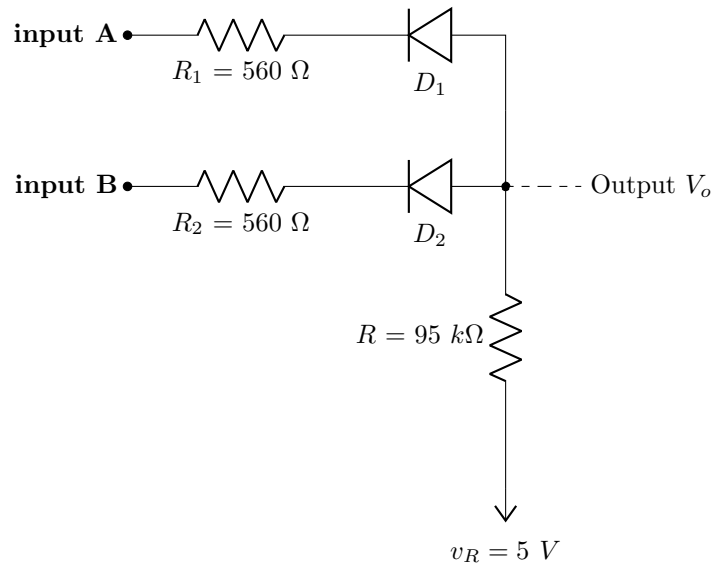


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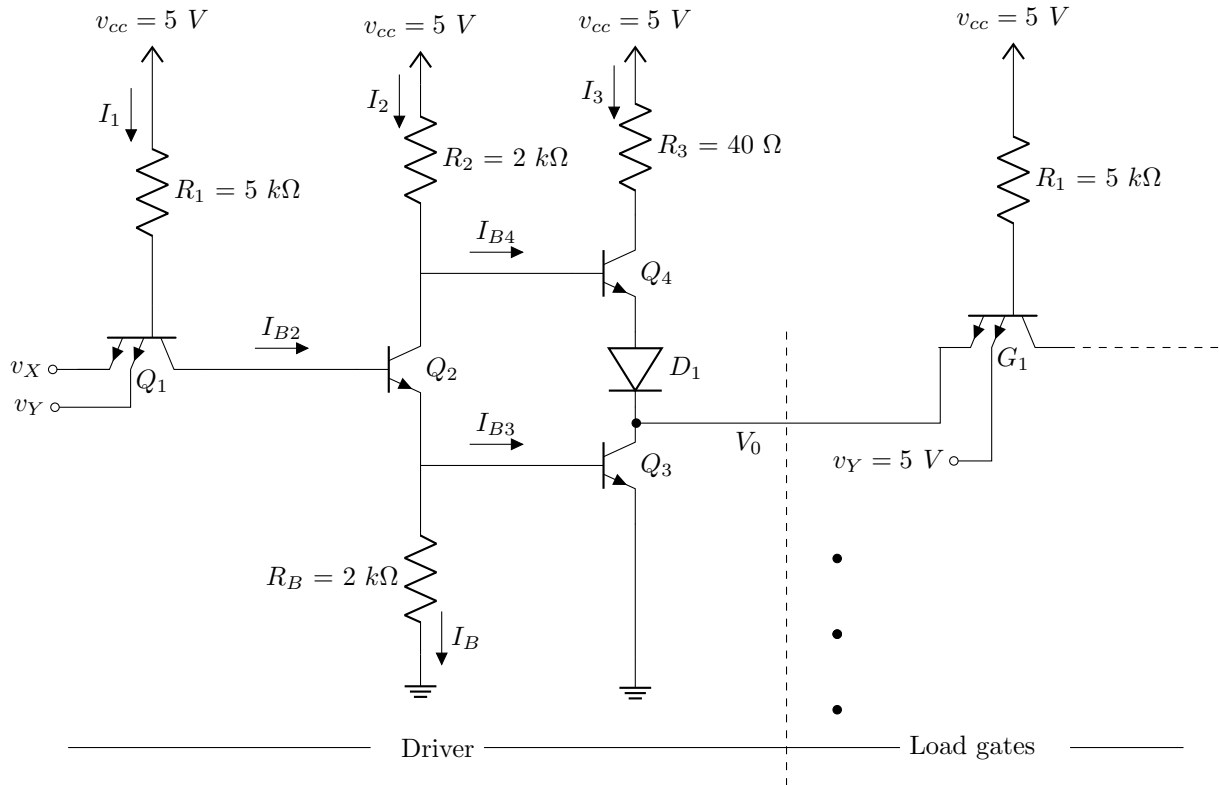


Figure 2

CO2	(a)	For fanout of $N = 8$, measure the output voltage V_o	[8]
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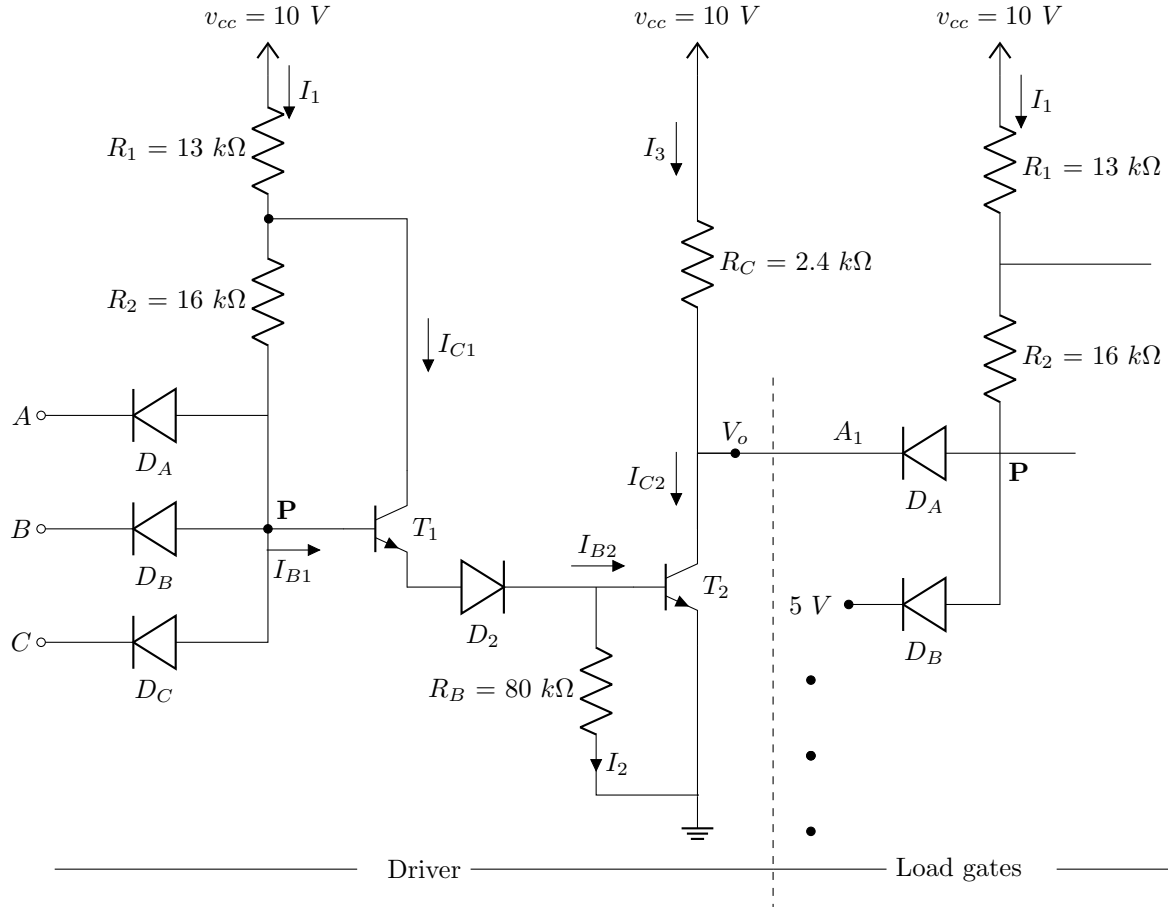


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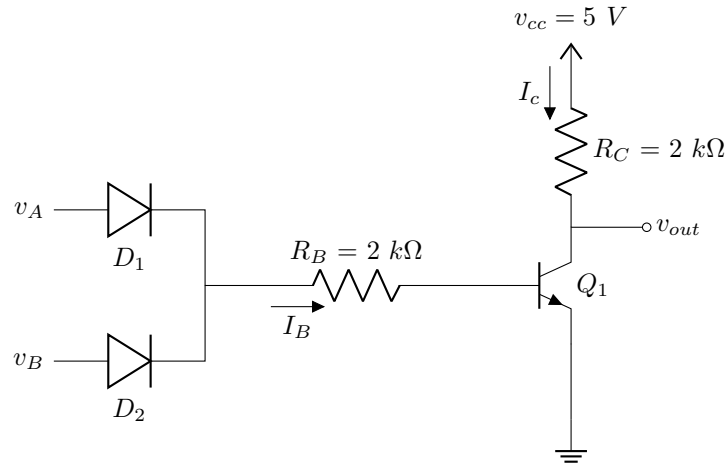


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